corresponding value of y on the y- axis, which will be the required interpolated value.

From the graph in Fig. 5.1 we find that for $x = 38$, the value of y is equal to 35.

5.2.3 Algebraic method

Newton's Gregory forward interpolation formula (or) Newton's forward interpolation formula (for equal intervals).

Do YOU KNOW?

The first two terms will give the linear interpolation and the first three terms will give a parabolic interpolation and so on.

Let $y = f(x)$ denote a polynomial of degree n which takes $(n+1)$ values. Let them be $y_0, y_1, y_2, \ldots y_n$ corresponding to the values $x_0, x_1, \ldots x_n$ respectively.

The values of x $(x_0, x_1, x_2, \ldots x_n)$ are at equidistant.

(i.e.) $x_1 = x_0 + h,\ x_2 = x_0 + 2h,\ x_3 = x_0 + 3h, \ldots, x_n = x_0 + nh$

Then the value of $f(x)$ at $x = x_0 + nh$ is given by

$$f(x_0 + nh) = f(x_0) + \frac{n}{1!}\Delta f(x_0) + \frac{n(n-1)}{2!}\Delta^2 f(x_0) + \frac{n(n-1)(n-2)}{3!}\Delta^3 f(x_0) + \ldots$$

(or) $$y_{(x=x_0+nh)} = y_0 + \frac{n}{1!}\Delta y_0 + \frac{n(n-1)}{2!}\Delta^2 y_0 + \frac{n(n-1)(n-2)}{3!}\Delta^3 y_0 + \ldots \text{ where } n = \frac{x - x_0}{h}$$

Newton's Gregory backward interpolation Formula.

Note

Newton's forward interpolation formula is used when the value of y is required near the beginning of the table.

In general Newton's forward interpolation formula not to be used when the value of y is required near the end of the table. For this we use another formula, called Newton's Gregory backward interpolation formula.

Then the value of $f(x)$ at $x = x_n + nh$ is given by

$$f(x_n + nh) = f(x_n) + \frac{n}{1!}\nabla f(x_n) + \frac{n(n+1)}{2!}\nabla^2 f(x_n) + \frac{n(n+1)(n+2)}{3!}\nabla^3 f(x_n) + \ldots$$

(or) $$y_{(x=x_n+nh)} = y_n + \frac{n}{1!}\nabla y_n + \frac{n(n+1)}{2!}\nabla^2 y_n + \frac{n(n+1)(n+2)}{3!}\nabla^3 y_n + \ldots \text{ when } n = \frac{x - x_n}{h}$$

Note

Newton's backward interpolation formula is used when the value of y is required near the end of the table.

Example 5.13

Using Newton's formula for interpolation estimate the population for the year 1905 from the table:

Year	1891	1901	1911	1921	1931
Population	98,752	1,32,285	1,68,076	1,95,670	2,46,050

Solution

To find the population for the year 1905 (i.e) the value of y at $x = 1905$.

Since the value of y is required near the beginning of the table, we use the Newton's forward interpolation formula.

$$y_{(x=x_0+nh)} = y_0 + \frac{n}{1!}\Delta y_0 + \frac{n(n-1)}{2!}\Delta^2 y_0 + \frac{n(n-1)(n-2)}{3!}\Delta^3 y_0 + \ldots$$

To find y at $x = 1905$ $\therefore x_0 + nh = 1905$, $x_0 = 1891, h = 10$.

$1891 + n(10) = 1905 \quad \Rightarrow n = 1.4$

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